

Iron

Fe (Fe I). Strongest classified transitions below 2 μm (NIST): 47 levels, 71 lines.

Fe⁺ ionisation limit 63737.70 cm⁻¹ = 7.90247 eV (156.893 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||er||J'\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A: Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

